

Atharva Jadhav

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Summary

AI Engineer with expertise in LLMs, deep learning, and production ML pipelines. Proven experience in model optimization, secure deployment, and cross-functional collaboration for AI-driven solutions.

Skills

- **AI/ML:** End-to-end ML system design, LLM integration, RAG pipelines, NLP/NER, transformer architectures, feature engineering, model evaluation, drift detection, statistical modeling, deep learning, multimodal workflows, performance optimization, responsible AI practices.
- **Tools & Algorithms:** Python, TensorFlow, PyTorch, Scikit-Learn, Keras, Pandas, NumPy, Spark, MLflow, Docker, REST APIs, CI/CD, Airflow ETL, gradient boosting, SQL, secure coding practices.
- **Cloud & Deployment:** AWS ML services, containerized deployment, scalable APIs, Linux, GitHub Actions.
- **Design & Prototyping:** System architecture design, workflow mapping, Figma, UI patterns, prototyping.

Projects

- Distributed LLM Training Pipeline (GPT-2 TinyLlama), Clemson SC** *Oct 2025 - present*
 - Engineered modular training system with DDP, FSDP, DeepSpeed-ZeRO achieving 4× GPU scaling efficiency gains.
 - Optimized multi-GPU throughput and convergence through systematic model configuration experimentation.
- Research Automation Platform, Clemson SC** *Aug 2025 - Oct2025*
 - Built automated LLM-powered pipeline for research paper retrieval, filtering, and structured data extraction.
 - Developed PDF-to-structured-data system converting scientific documents to CSV/JSON for downstream analytics.
- Multimodal Legal Document Analyzer, Clemson SC** *Jun 2025 - Aug 2025*
 - Created production ML pipeline with OCR, NER, transformers, scheduled training, MLflow versioning, drift checks.
 - Deployed secure containerized inference API with MongoDB integration and CI/CD for compliant document processing.
- Brain Age Estimation via MRI Set Learning, Clemson SC** *Aug 2024 - Dec 2024*
 - Built PyTorch pipeline (ResNet + DeepSets) for multi-MRI prediction with optimized data loading and GPU batching.
 - Implemented containerized API with MLflow tracking, monitoring, and automated testing for reliable deployment.
- Heart Disease Prediction Using Machine Learning, Pune India** *Jun 2021 - May 2022*
 - Engineered end-to-end ML pipeline for cardiovascular risk prediction with hyperparameter tuning and validation.
 - Deployed Flask-based inference API with MLflow experiment tracking ensuring reproducible training workflows.

Work Experience

- Machine Learning Research Volunteer – NanoBio Mechanics and Manufacturing Lab Clemson SC** *Oct 2025 - Present*
 - Developed ML models for filtration optimization using morphology features from SEM imaging and experimental data.
 - Built data processing pipelines for feature extraction enabling data-driven materials science optimization insights.
- Data Science Intern – CodeBita Technology, Pune** *Jul 2022 - Jul 2023*
 - Built predictive ML models and analytics pipelines supporting business intelligence and operational decision-making.
 - Automated data preprocessing workflows and created interactive dashboards for stakeholder metric visualization.

Education

- **M.S in Computer Science**, Clemson University, SC, USA (*May 2025*) – GPA: 3.73/4.0
- **B.E in Computer Engineering**, SPPU, India (*June 2022*) – GPA: 3.46/4.0

Certifications

- TCS MasterCraft DevPlus - TCS (Development and engineering fundamentals)
- Accenture Developer Program - Forage (software engineering foundations)
- Agile Project Management - Udemy (Scrum, sprint planning, backlog execution)

Leadership and Involvement

- Research collaboration with cross-functional teams including materials scientists and domain experts
- Peer code review participation ensuring quality standards and knowledge sharing across projects